Robin Hur

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Education

University of Virginia, School of Engineering and Applied Science **Charlottesville, VA**
B.S. in Mechanical Engineering, GPA: 3.75 **Expected May 2027**

- **Honors/Awards:** Dean's List, Bridge Scholar
- **Relevant Coursework:** Manufacturing Process Technology, Mechatronics, Strength of Materials, Statics, Dynamics, Fluid Mechanics, Thermodynamics, Materials Science

Skills

- **DFM & Mechanical Design:** SOLIDWORKS (CSWA), Fusion 360, Maya, Tecplot 360, GD&T, Technical Drawings
- **Process Data Analysis & Controls:** Python, MATLAB, Simulink, Linux, ANSYS, CFD, High-Performance Computing
- **Certifications & Languages:** English (native), Korean (fluent), French (intermediate)

Experience

Research Assistant, *UVA Dept. of Materials Science and Engineering* **Feb 2026 – Present**

- Developed closed-loop feedback control in Python to dynamically regulate N_2 flow and relative humidity.
- Rapidly architected an error-free Python control interface via AI-assisted workflows, cross-referencing legacy humidity software and THProbe defaults to maintain relative humidity within $\pm 2\%$ over 20+ hours.
- Advanced environmental cracking test setups for high-strength aerospace aluminum alloys supporting research sponsored by NASA, Lockheed Martin, and the F-35 program.

Research Assistant, *UVA Dept. of Mechanical and Aerospace Engineering* **May – August 2026**

- Executed CFD simulations and numerical design optimization on bio-inspired underwater robotic systems, evaluating part-level geometric modifications to improve propulsion efficiency and hydrodynamic performance.
- Set up automated Python and parallel processing scripts on the university HPC cluster to process 3D flow field data, accelerating iterative design reviews and thrust/drag trade studies.
- Presented weekly technical reports and design recommendation packages to cross-functional engineering teams to guide hardware iteration.

Research Assistant, *UVA Dept. of Chemical Engineering* **Aug 2024 – May 2026**

- Analyzed atomic-level assembly mechanisms during hydrogel gelation using all-atom molecular dynamics (MD) simulations in AMBER, identifying key structural variables affecting mechanical yield.
- Built automated Python pipelines with Git version control to process simulation trajectories in VMD, managing data architecture for 10+ TB of test data across 100+ HPC configurations.
- Co-authored *A Methodological Approach to Relate Pentapeptide Sequence, Atomistic Self-Assembly, and Gelation Behavior*, ACS Biomaterials Science & Engineering.

Teaching Assistant, *UVA Dept. of Electrical and Computer Engineering* **Jan 2026 – May 2026**

- Supported undergraduate students during embedded systems laboratories and office hours, grading 450+ students' worksheets alongside peers.
- Collaborated with faculty to evaluate technical assignments and refine lab instruction documentation for quality and clarity.

Projects

Mechatronics and Robotics Society, *Mechanical Subteam Member* **Aug 2025 – Dec 2025**

- Modeled multi-part CAD assemblies in SOLIDWORKS for a NASA Lunabotics excavation robot, applying DFM/DFA principles to ensure modular assembly, structural rigidity, and tight-tolerance component fitment.
- Evaluated granular flow and excavation subsystem mechanisms, identifying stress concentration areas and failure modes to optimize material handling efficiency in simulated lunar soil.

Potential Energy Propulsion Racer, *Mechanical Lead* **Nov 2025**

- Designed and fabricated a 3D-printed vehicle transmission and frame using SOLIDWORKS, conducting tolerance analysis to guarantee proper gear mesh clearances, press-fit tolerances, and low friction.
- Curated complete engineering documentation including multi-part assembly drawings, a comprehensive Bill of Materials, and mechanical trade-off analyses.